

SANJAY KUMAR G

AI/ML Engineer | Data Science | GenAI & LLM Applications | Python Developer
Chennai, Tamil Nadu | +91 9361039676 | sanjaykumar3092004@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science fresher specializing in AI/ML and Data Science, with hands-on experience building RAG systems, fine-tuning LLMs (LoRA/QLoRA), and developing real-time computer vision models achieving 90%+ accuracy. Proficient in Python, SQL, TensorFlow, LangChain, and Hugging Face Transformers, with a strong foundation in statistics and ML algorithms. Seeking a Data Scientist / AI-ML Engineer role to build and deploy production-grade ML and GenAI solutions.

TECHNICAL SKILLS

Languages: Python, Java, SQL

ML / Data Science: Scikit-learn, Feature Engineering, Model Tuning, Statistics & Hypothesis Testing, CNN, NLP

GenAI / LLMs: LangChain, RAG Pipelines, LoRA/QLoRA Fine-Tuning, Hugging Face Transformers & PEFT, Ollama, FAISS, Vector Databases, Prompt Engineering

Frameworks & Tools: TensorFlow, OpenCV, YOLO, Flask, REST API, NumPy, Pandas, Streamlit, Git, Postman, Jupyter, Google Colab

Databases: MySQL (Joins, Indexing, Query Optimization)

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

PROJECTS

AI Knowledge Assistant using RAG — *LangChain | Ollama (Phi-3) | FAISS | HuggingFace Embeddings | Streamlit*

- Developed an AI-powered Knowledge Assistant using LangChain, Ollama (Phi-3), FAISS, and HuggingFace Embeddings to answer user queries from uploaded PDF documents through a Retrieval-Augmented Generation (RAG) pipeline.
- Implemented PDF parsing, text chunking, semantic search, vector database indexing, and context-aware question answering with an interactive Streamlit interface.
- Built a modular architecture with document loading, text splitting, vector storage, retrieval, and LLM integration, enabling efficient and scalable document-based AI assistance.

AI Interview Coach — LLM Fine-Tuning — *Python | Hugging Face Transformers | PEFT (LoRA) | Phi-3*

- Fine-tuned Phi-3 Mini (3.8B) using LoRA and 4-bit quantization on 800+ interview Q&A pairs to generate role-specific technical interview questions and ideal answers via an interactive Gradio interface.

Traffic Management System — ML & Computer Vision — *Python | YOLOv5 | OpenCV | Raspberry Pi*

- Engineered a real-time vehicle detection system using YOLOv5 and OpenCV, achieving 90%+ accuracy and reducing simulated congestion by 25%.
- Built a scalable edge-to-cloud pipeline integrating Raspberry Pi with cloud processing for sub-second real-time inference.

Sign Language Recognition System — Deep Learning — *Python | TensorFlow | CNN | OpenCV*

- Implemented a real-time hand-gesture classifier using CNN and TensorFlow with 90%+ accuracy across 26 ASL signs.
- Optimized the inference pipeline to cut end-to-end latency by 30% for smooth real-time performance.

Student Management System — Backend — *Python | Flask | MySQL | REST API*

- Built a backend system with 10+ RESTful API endpoints managing 1,000+ student records with full CRUD support.
- Optimized MySQL queries via indexing and normalization, improving retrieval performance by 40%.

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML)

2022 – 2026

Vels Institute of Science, Technology & Advanced Studies, Chennai | CGPA: 7.8/10

CERTIFICATIONS & ACHIEVEMENTS

- Generative AI — Imarticus Learning (2025)
- Machine Learning A–Z — VOIS (2024)
- Cyber Security Associate — Futurecalls Technology (2023)
- Paper Presentation Runner-Up — GIST National Level Symposium (2025)